

Haett Churn Prediction API 1.0.0 OAS 3.1

[/openapi.json](#)

ML-powered churn prediction for the Haett meal delivery platform. Predicts user churn probability within the next 30 days and provides actionable business recommendations.

System



GET

/health

Health Check

⌵

Health check endpoint.

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
'http://localhost:8000/health' \
-H 'accept: application/json'
```

⌵

Request URL

```
http://localhost:8000/health
```

Server response

Code	Details
200	Response body <pre>{ "status": "healthy", "model_loaded": true,</pre>

Code

Details

 **Download**

```
"version": "1.0.0"
}
```

Response headers

```
content-length: 58
content-type: application/json
date: Tue, 14 Jul 2026 15:33:46 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
{
  "status": "string",
  "model_loaded": true,
  "version": "string"
}
```

GET / Root

Root endpoint with API info and links.

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:8000/' \
  -H 'accept: application/json'
```



Request URL

http://localhost:8000/

Server response

CodeDetails

200

Response body

```
{
  "service": "Haett Churn Prediction API",
  "version": "1.0.0",
  "docs": "/docs",
  "health": "/health",
  "predict": "/predict"
}
```

Download

Response headers

```
content-length: 113
content-type: application/json
date: Tue,14 Jul 2026 15:33:53 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

Prediction



POST /predict Predict Churn



Predict churn probability for a single user.

Takes user features and returns:

- churn_probability: probability of churning within next 30 days
- risk_level: Low (< 30%), Medium (30-60%), or High (> 60%)
- business_recommendation: actionable retention recommendation

Add ?explain=true to also receive the top 5 feature contributions via SHAP.

Parameters

[Cancel](#)[Reset](#)

Name	Description
explain	Include SHAP feature explanations in the response
boolean	false
(query)	

Request body required

[application/json](#)[Edit Value](#) | [Schema](#)

```
{
  "user_id": 1,
  "days_since_last_order": 60,
  "tenure_days": 75,
  "total_orders": 5,
  "std_days_between_orders": 15,
  "orders_last_30_days": 0,
  "avg_order_value": 28,
  "avg_rating": 2.1,
  "coupon_usage_count": 2,
  "coupon_usage_rate": 0.4,
  "n_plan_changes": 3,
  "monthly_price": 29.99,
  "subscription_tenure_days": 30,
  "avg_app_logins": 0.5,
  "avg_meals_skipped": 3,
  "total_support_tickets": 8,
  "age": 22,
  "age_group_code": 0
}
```

[Execute](#)[Clear](#)

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8000/predict?explain=false' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "user_id": 1,
    "days_since_last_order": 60,
    "tenure_days": 75,
```

```
"total_orders": 5,
"std_days_between_orders": 15,
"orders_last_30_days": 0,
"avg_order_value": 28,
"avg_rating": 2.1,
"coupon_usage_count": 2,
"coupon_usage_rate": 0.4,
"n_plan_changes": 3,
"monthly_price": 29.99,
"subscription_tenure_days": 30,
"avg_app_logins": 0.5,
"avg_meals_skipped": 3,
"total_support_tickets": 8,
"age": 22,
"age_group_code": 0
},'
```

Request URL

http://localhost:8000/predict?explain=false

Server response

Code	Details
200	Response body <pre>{ "user_id": 1, "churn_probability": 0.2269, "risk_level": "Low", "business_recommendation": "No action needed. User is at low risk of churning.", "explanations": null }</pre> Download Response headers <pre>access-control-allow-credentials: true access-control-allow-origin: http://localhost:8000 content-length: 158 content-type: application/json date: Tue,14 Jul 2026 15:38:07 GMT server: uvicorn vary: Origin</pre>

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema <pre>{ "user_id": 0, "churn_probability": 1, "risk_level": "Medium", "business_recommendation": "string", "explanations": [</pre>	No links

Code	Description	Links
	<pre>{ "feature": "string", "value": 0, "impact": 0 }</pre>	
422	Validation Error	No links
	Media type application/json	
	Example Value Schema	
	<pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string", "input": "string", "ctx": {} }] }</pre>	

POST /predict/batch Predict Churn Batch

Schemas

BatchPredictionRequest > Expand all object

ChurnPredictionRequest > Expand all object

ChurnPredictionResponse > Expand all object

FeatureExplanation > Expand all object

HTTPValidationError > Expand all object

HealthResponse > Expand all [object](#)

ValidationError > Expand all [object](#)